

Qatar-10b

R-0098

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_231003 · created 2026-07-16T02:09:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460220.6554 +/- 0.008 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.03
Transit depth (Rp/Rs)^2	0.63 +/- 0.47 [%]
Orbital Inclination (inc)	89.77 +/- 2.54
Airmass coefficient 1 (a1)	1.219 +/- 0.016
Airmass coefficient 2 (a2)	-0.128 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	None
Optimal Aperture	2.93
Optimal Annulus	7.95
Transit Duration (day)	0.104 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.079 ± 0.03 vs 0.1265 (deviation 1.58σ)
Duration fit vs published	0.104 d vs 0.1154 d (deviation 1.48σ)
Mid-time O–C	2.2 min
Depth SNR	2.6
Residual scatter	1.3 %

Light curve

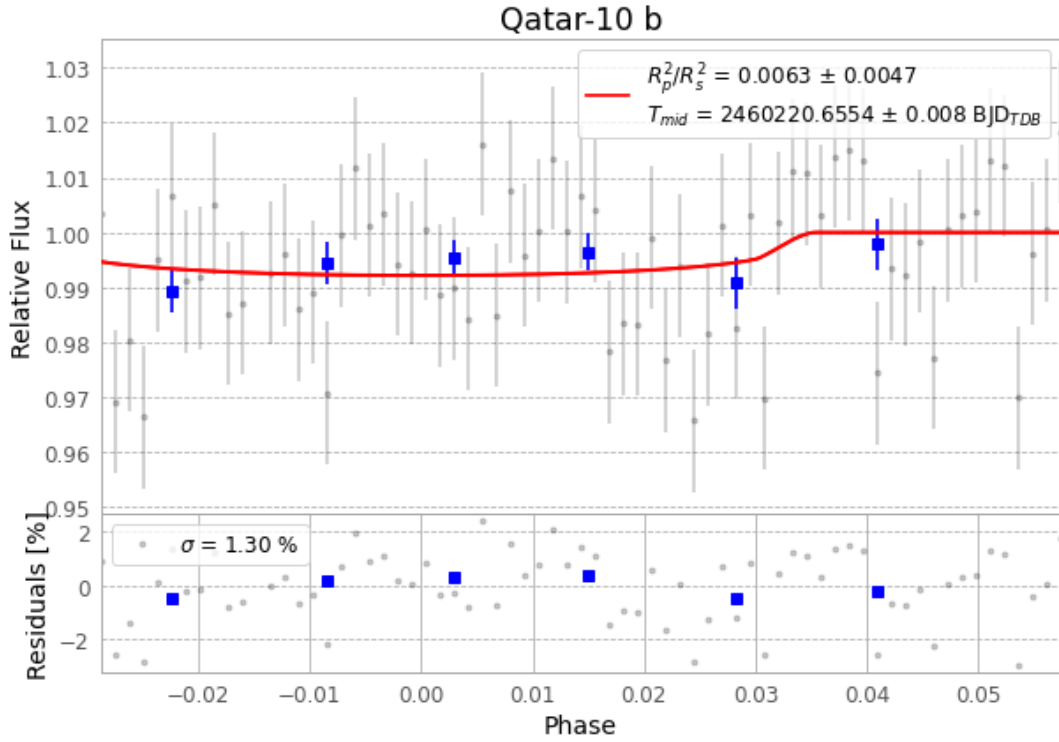


Figure 1 — FinalLightCurve_Qatar-10 b_2023-10-02.png (SHA-256 fb9789a5ce1836dc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0579afe271895af3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

69 FITS frames (45 MB), Qatar-10231003023315.FITS → Qatar-10231003055715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-231003.log (ff600306e4d4...), FinalLightCurve_Qatar-10 b_2023-10-02.png (fb9789a5ce18...), FinalLightCurve_Qatar-10 b_2023-10-02.csv (b1898e4552f0...)

Compute resources

Wall time 200.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 02:09Z.